SERVER:
OPEN CHIPLET
ECONOMY

The Open Chiplet Economy: Progress, Challenges, and the Road Ahead

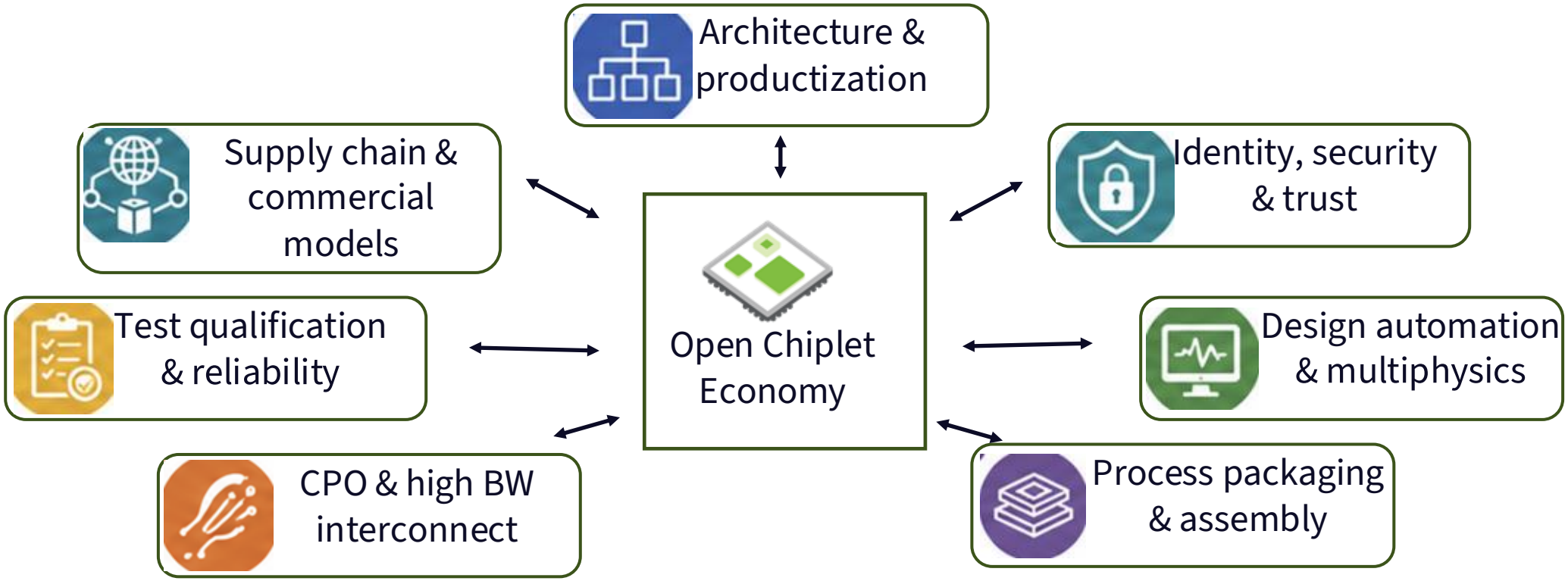


Jawad Nasrullah, *Palo Alto Electron*

Anu Ramamurthy, *Microchip*

Mark Knight, *Arm*

Optimizing Silicon for AI - Open Chiplet Economy



Each layer has an owner. No organization owns the complete open-system outcome.

Introduction to Open Chiplet Economy (OCE)

Open collaboration and standardization for multi-vendor chiplet systems

EDA Tools & Flows

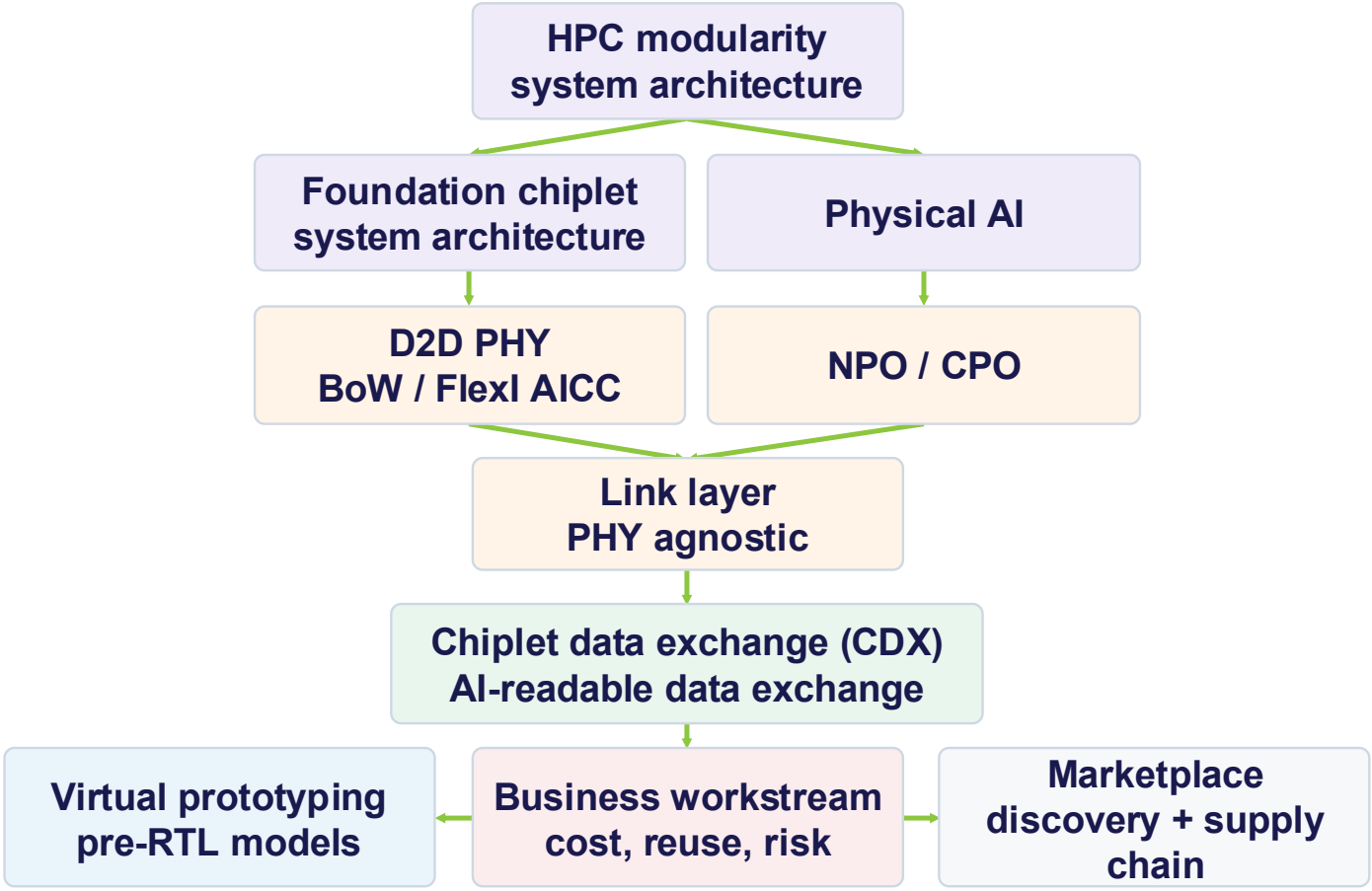
Exchange design data and models.
Evaluate integration before RTL.

Open Interfaces

Connect chiplets through open physical, link and management interfaces.

Ecosystem Standards

Define system architecture for interoperability and reuse.



2026 Progress Report

FCSA Workstream

Initial Release 1.0.0

February 2026

NEW Release 1.1.0

Approved & published this week

(See 1:10 PM session for details)

System Workstream

CMI over I3C Completed

Waiting for OCP approval

CMI over UCIe In Review

starting OCP internal reviews

CMI over BOW Initial Draft

Dedicated SCMI API Initial Draft

starting initial reviews

LSI Workstream

Interconnect Standard

Standardizing lightweight SerDes interconnect between die

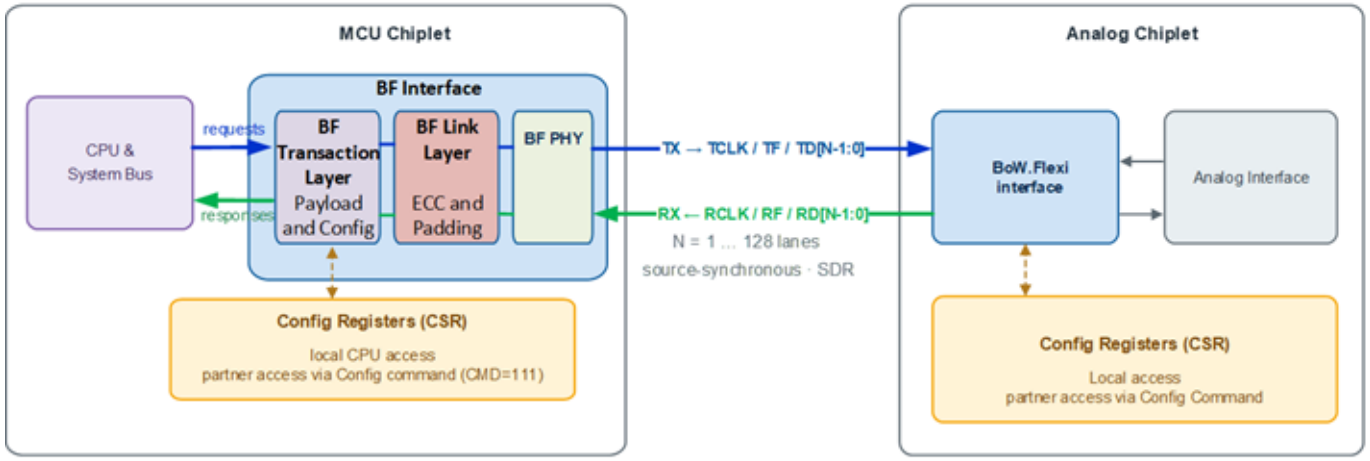
Ecosystem Contribution

Credo and their partners contribution to the OCE - Draft

2026 Progress Report

BoW Flexi 1.0 release - September 2026

- Low overhead, low cost and power D2D
- Target Applications: MCU, Sensors, Edge AI, IoT



VeriSilicon MICROCHIP eliyan YorChip Infineon

BoW.Flexi V1.0.0

Contributor: VeriSilicon Microelectronics, Microchip Technology, Inc., Eliyan Corporation, YorChip, Infineon Technologies

Family: Server

Project: Server > Open Chiplet Economy > Bunch of Wires

Release Year: 2026

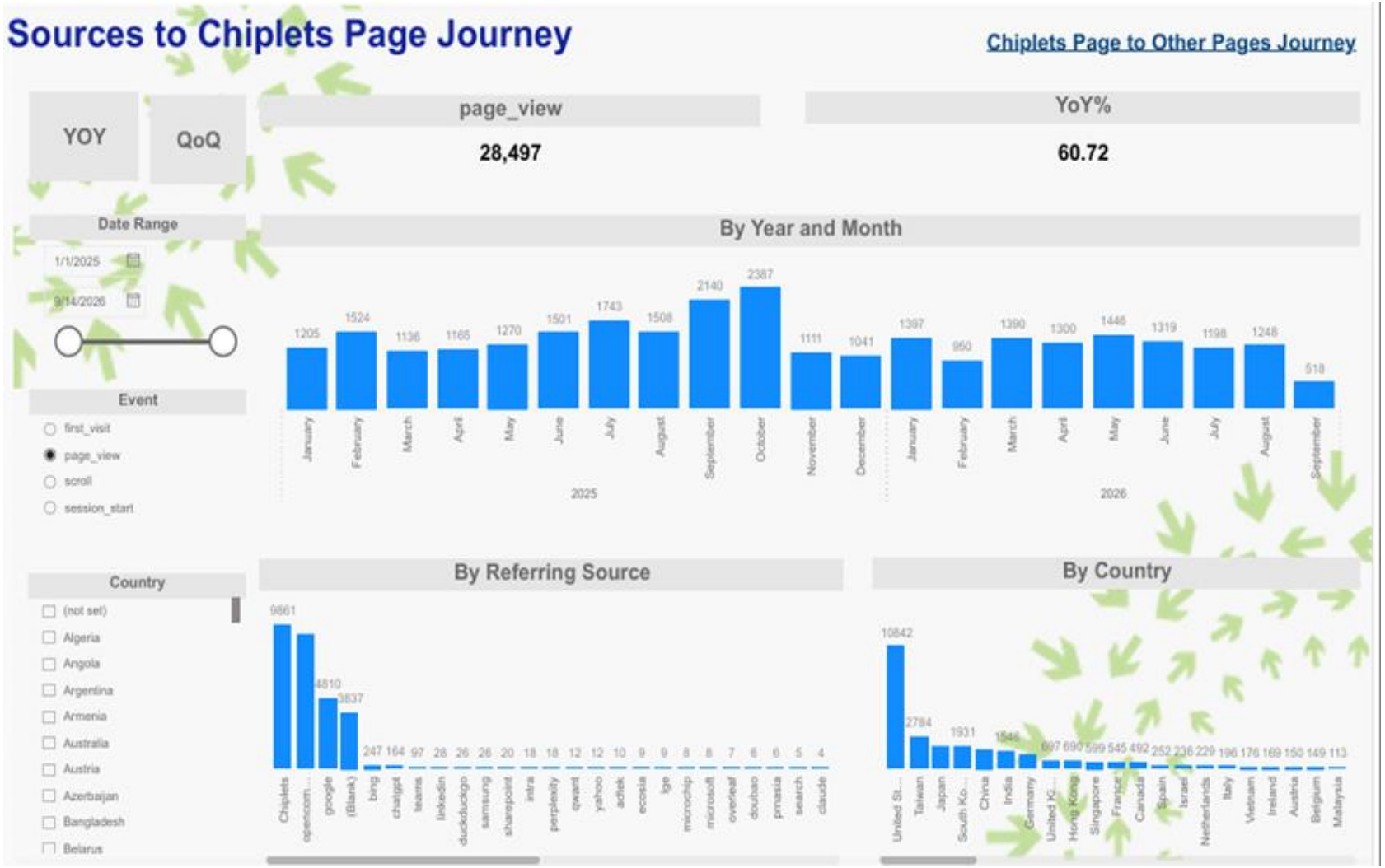
Chiplet marketplace progress

28,497 page views

60% YoY

Peak monthly engagements ~ 2837

Broad reach across the continents



New volunteer contributors - thank you





















Focus areas for 2027

2027 should be the year the chiplet industry advances from defining standards to demonstrating interoperability:

- ✅ Make interoperability verifiable
- ✅ Make chiplet reuse commercially attractive

System level interoperability and Security

- Continue to fully specify the contract between chiplets

BoW flexi D2D - next generation

- Incorporating feedback from real implementations
- Look into 3D stacking

Chiplet design exchange

- Deliver Proof of Concept with partners

Physical AI

- Address unique requirements of automotive and robotics

Investigating a New Business Workstream

New business models and relationships are needed to realize benefits of a chiplet ecosystem.

Opportunity to Reduce Commercial Barriers

Improve Time to Market
Accelerate adoption and deployment cycles across vendors.

Reduce Risk of Failure & Costly Re-work
Establish clear standards and alignment early.

Simplify CDX Participation
Streamline Chiplet Design eXchange integration.

Target Workstream Deliverables

Product Development Project Structure
Example frameworks for multi-vendor projects.

Case Studies
Real-world scenarios and operational learnings.

Model Agreements
Standardized templates to lower legal friction.

Today in the Open Chiplet Economy

What will it take to build deployable systems from multi vendor chiplets?

8:00 AM Opening overview Progress this year, remaining barriers, and priorities ahead

WHERE TO GO DEEPER TODAY

System integration	8:20 AM Custom HBM as a system	9:10 AM Interoperable AI systems
Die-to-die interfaces	11:15 AM BoW Flexi 1.0	12:30 PM Transaction & link layers
Security and trust	9:45 AM Caliptra root of trust	1:10 PM Common security architecture
Design and simulation	10:30 AM Virtual chiplet ecosystems	1:55 PM Federated design & simulation
Physical AI	2:45 PM New FCSA workstream	3:10 PM Physical AI panel

Thursday, October 15 • Room 211 • Selected sessions; titles shortened

Call to Action

Four ways to participate in an Open Chiplet Economy

1. Contribute to a new or existing workstream
2. Join meetings and provide feedback
3. Implement to OCE specification
4. Ask your suppliers to build to OCE specifications

https://scnv.io/37ka

Thank You!

